XLRON: A Framework for Hardware-Accelerated and Differentiable Simulation of Optical Networks

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XLRON is a high-performance simulation library for optical networks. Designed to enable research on reinforcement learning (RL) for resource allocation, XLRON also supports gradient-based optimization and features a closed-form Gaussian noise (GN) physical layer model that supports inter-channel stimulated Raman scattering (ISRS) with Nyquist digital subchannels and coherent noise accumulation. Among open-source optical network simulation libraries, XLRON provides both the most comprehensive feature set and the highest computational performance on CPU and GPU, enabling rapid experimentation. We highlight the capabilities of XLRON by comparison with other optical network simulation libraries, demonstrating RL training completion up to 1,750× faster, and use its physical layer model to recreate a recent record throughput experiment from the literature, matching the reported per-band generalized OSNR to within 0.6 dB and the achieved throughput to within 0.6%. We use XLRON's fully differentiable simulation to optimize routing and spectrum allocation by gradient descent, allocating more services than a first-fit heuristic baseline. We anticipate that XLRON's combination of high computational throughput, differentiable simulation and an ISRS GN physical layer model in one framework, and its ease of use for both humans and AI coding agents will make it a valuable tool for the optical networking research community.

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1. INTRODUCTION

Optical networks are complex cyber-physical systems covering many orders of magnitude of time and distance scales, ranging from trans-oceanic to data center networks. These now face increasing bandwidth demands and require significant ongoing capital and operational expenditure. Consequently, optimization of optical network planning and operations is highly desired. Access to live operational networks or digital twins [10] is limited, therefore simulation is an essential method to enable experimentation and refinement of optimization techniques.

Historically, researchers in this area have developed simulation programs ad-hoc for a particular project, or re-used closed source code from within their organizations. In recent years, following the trend in open source software and with the aim of improving research productivity and reproducibility, several optical network simulation libraries have been published [1–7], with the objectives of supporting development of resource (routing and spectrum) allocation algorithms, lightpath SNR estimation, and training of machine learning models for a range of network optimization tasks and even network design and planning.

While these tools improve on the inefficiency and opacity of ad-hoc and closed-source software development, the proliferation of new simulation frameworks has introduced a level of complexity in choosing a tool for a particular use case, and differing implementation details can lead to vastly different simulation outcomes across frameworks [11]. The problems of tool choice and inconsistent implementations may be compounded by the rise of agentic coding tools. These significantly reduce the barriers to software development but increase the risk of further ecosystem fragmentation and issues with reliability and interpretability of results.

The XLRON framework addresses the problem of tool choice by providing a reliable, easy-to-use framework that supports the majority of optical network simulation workloads with high computational performance and an extensive feature set (Table 1). Its emphasis is high-throughput simulation under any of the standard operational paradigms — static, incremental or dynamic traffic — at a configurable degree of physical-layer detail, from pure resource allocation to a full ISRS GN model evaluated for every lightpath. It is therefore a gen-

Feature	GNPy [1]	DeepRMSA [2]	RSA-RL [3]	ORL-Gym [4]	ON-Gym [5]	FUSION [6]	Flex Net Sim [7]	XLRON
RL training	×	✓	✓	✓	✓	✓	Ext.	✓
Invalid action masking	×	×	✓	×	×	×	×	✓
Physical layer model	GN	×	×	×	GN	Partial	×	EGN
ISRS	Partial	×	×	×	×	×	×	✓
Nyquist subchannels	×	×	×	×	×	×	×	✓
Multi-band	Partial	×	×	×	✓	Partial	Ext.	✓
Differentiable simulation	×	×	×	×	×	×	×	✓
Hardware acceleration	×	×	×	×	×	×	×	GPU
GUI	✓	×	×	×	×	✓	×	✓
Scriptable CLI	✓	×	×	×	×	✓	×	✓

Table 1. Feature comparison of open-source optical network simulation libraries, assessed as of July 2026. "Ext." denotes capabilities that are not provided natively but have been demonstrated by the library authors via external extensions, in this case for Flex Net Sim: a deep RL adapter [8] and a space-division-multiplexing extension with inter-core crosstalk modeling [9].

eral simulation platform rather than a single-purpose one: the throughput suits machine-learning workloads without being restricted to them, which is why XLRON can be compared with libraries aimed at detailed physical-layer calculation (GNPy), at heuristic resource-allocation algorithms (Flex Net Sim), and at reinforcement learning (RL) training (ON-Gym). XLRON is implemented entirely in JAX [12], which provides just-in-time (JIT) compilation and automatic vectorization across CPU and GPU hardware; by compiling and vectorizing the entire simulation pipeline, it executes thousands of independent simulation instances in parallel on a single device, yielding throughput several orders of magnitude higher than pure-Python simulators while remaining within the Python ecosystem familiar to most researchers. A preliminary version was introduced in [13]; the physical layer model, differentiable simulation, interfaces and all benchmarks reported here are new.

The remainder of this paper describes the XLRON contributions through a series of comparisons and case studies:

- a feature comparison with seven other open-source optical network simulation libraries (Section 2), and computational performance benchmarks demonstrating orders of magnitude faster RL training than existing frameworks (Section 3);
- a closed-form Gaussian noise (GN) physical layer model accounting for inter-channel stimulated Raman scattering (ISRS), Nyquist digital subchannels, and distributed Raman amplification (DRA), validated against a recent record-throughput C+L-band transmission experiment (Section 4);
- the first fully differentiable optical network simulator, which uses straight-through gradient estimators [14] to enable gradient-based optimization of discrete resource allocation decisions (Section 5); and
- graphical and command-line interfaces designed for both interactive use and integration with large language model (LLM)-based coding agents, supported by a comprehensive suite of unit tests (Section 6).

Section 7 summarizes and highlights the significance of the results.

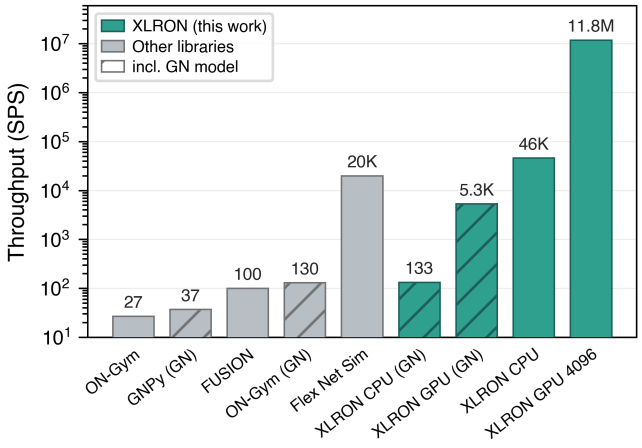


Fig. 1. Simulated steps per second (SPS) of the surveyed libraries on the 44-directed-link NSFNET topology with 100 frequency slot units (FSU) per link (KSP-FF, $k = 5$, distance-adaptive modulation formats). Hatched bars include a physical-layer GN model in the allocation loop. All bars are single-core figures except the two marked XLRON GPU. FUSION, ON-Gym (excluding GN), and Flex Net Sim values are computed from the wall-clock times in Table 1 of [7], which does not state its benchmark hardware; the ON-Gym GN value derives from the per-request processing time in [5] for one core of an AMD Ryzen Threadripper 3960X. GNPy and XLRON values are measured here on an Apple M1 Pro (CPU) and NVIDIA H100 (GPU).

2. COMPARISONS OF OPTICAL NETWORK SIMULATION LIBRARIES

Table 1 summarizes the features of the eight open-source optical network simulation libraries considered in this comparison. We briefly describe each library below and highlight the key differences, then compare their simulation throughput measured in steps per second (SPS), where each step is a traffic request allocation.

GNPy [1, 15] is the de-facto standard for vendor-agnostic quality-of-transmission estimation, maintained under the Telecom Infra Project for route planning in mesh optical networks.

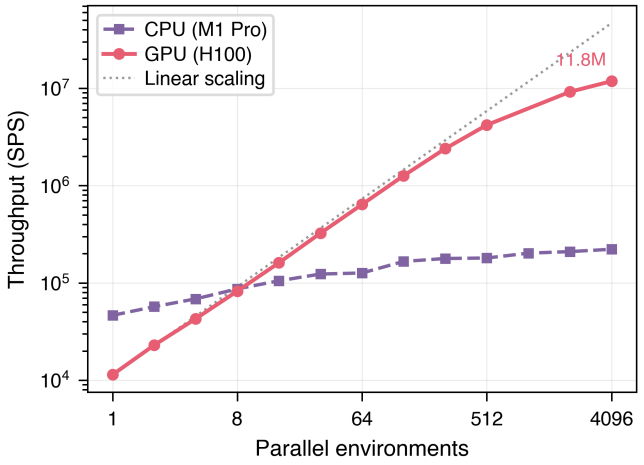


Fig. 2. XLRON RMSA simulation steps per second (SPS) vs. number of parallel environments on GPU and CPU. The dotted line indicates ideal linear scaling.

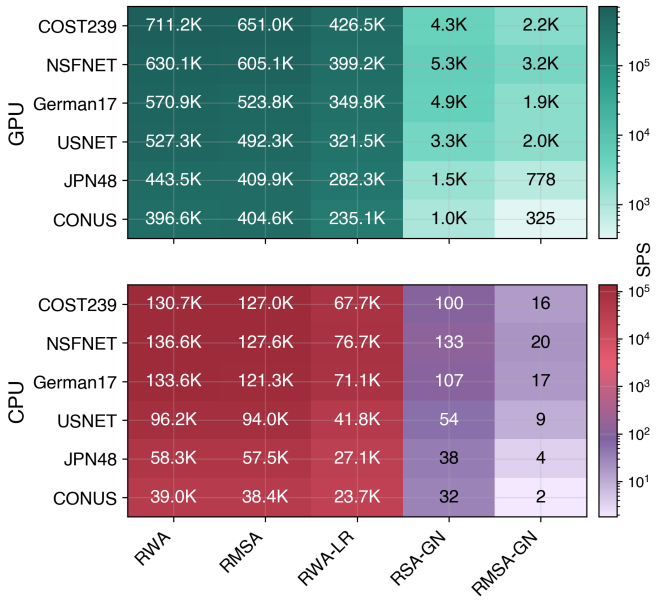


Fig. 3. XLRON throughput (SPS) across six topologies and five environment types on GPU (top) and CPU (bottom). RWA, RMSA and RWA-LR use 64 parallel environments; RSA-GN and RMSA-GN use one.

It targets detailed, equipment-level modeling of production networks rather than the high-throughput algorithm development XLRON is built for: it does not simulate dynamic allocation of traffic requests, and its per-lightpath physical-layer calculation (benchmarked in Section 3) is impractical for workloads of millions of steps.

DeepRMSA [2] was among the first works to apply deep RL to the routing, modulation, and spectrum assignment (RMSA) problem, and its open-source implementation has been adopted and extended by several subsequent works [16, 17]. RL is attractive for dynamic RMSA on three grounds: a trained policy returns an allocation in a single forward pass, so decision time is short and independent of network state; it can learn structure in the traffic and topology that a hand-designed heuristic does not exploit, and so accept more services; and its reward can express objectives beyond blocking — cost, energy consumption, or a weighted combination — without redesigning the allocation algorithm.

RSA-RL [3] applies invalid action masking [18] to the routing and spectrum assignment (RSA) problem, constraining the agent to feasible allocations and improving RL sample efficiency [19]. It is otherwise limited in scope, with no physical layer model or path-dependent modulation format selection.

Optical-RL-Gym (ORL-Gym) [4] provides OpenAI Gym-compatible environments for optical network resource allocation with RL, supporting dynamic RSA and RMSA, and has been extended to multi-band scenarios [20].

Optical-Networking-Gym (ON-Gym) [5] is the successor to ORL-Gym and improves simulation performance by implementing computationally intensive per-lightpath SNR calculations with a closed-form Gaussian Noise (GN) model in Cython, for quality-of-transmission (QoT)-aware resource allocation, and supports multi-band operation.

FUSION [6] provides an actively maintained, GUI-based environment supporting both RL training and heuristic evaluation, with partial physical layer modeling and multi-band support.

Flex Net Sim [7] is a recently released C++ library for high-performance, event-driven simulation of resource-allocation heuristics in elastic optical networks (EONs), with a modular architecture that users can extend. It is a highly optimized single-core engine for pure-heuristic spectrum allocation, with no native physical layer model or RL training, though user extensions have added space-division multiplexing with inter-core crosstalk [9] and a deep-RL wrapper [8].

XLRON provides the most comprehensive feature set among the libraries surveyed: it is the only one supporting every capability in Table 1, and additionally estimates upper bounds on the capacity of a given topology and traffic scenario [11, 21]. These serve distinct tasks — invalid action masking and GPU acceleration for RL training, the ISRS GN model for QoT-aware allocation, differentiability for gradient-based design — and with the performance of Section 3 make XLRON a unified framework for optical network simulation research.

A. Physical layer models

Quality-of-transmission-aware resource allocation has received growing attention, with several studies incorporating physical layer impairment models into RL-based RSA and RMSA [22–25]. Among the surveyed libraries only GNPy and ON-Gym ship a validated closed-form physical layer model comparable to XLRON; the others provide none. Neither, however, combines inter-channel stimulated Raman scattering (ISRS), distributed Raman amplification (DRA), Nyquist digital subchannels, and a modulation-format-dependent enhanced GN (EGN) correction in a single model, as XLRON does [26, 27]. XLRON is also the only surveyed library designed for RL training that re-evaluates the SNR of every active lightpath after each allocation, capturing the ISRS tilt and nonlinear-interference changes that later allocations induce on shared spans. Its physical layer model is evaluated in detail in Section 4.

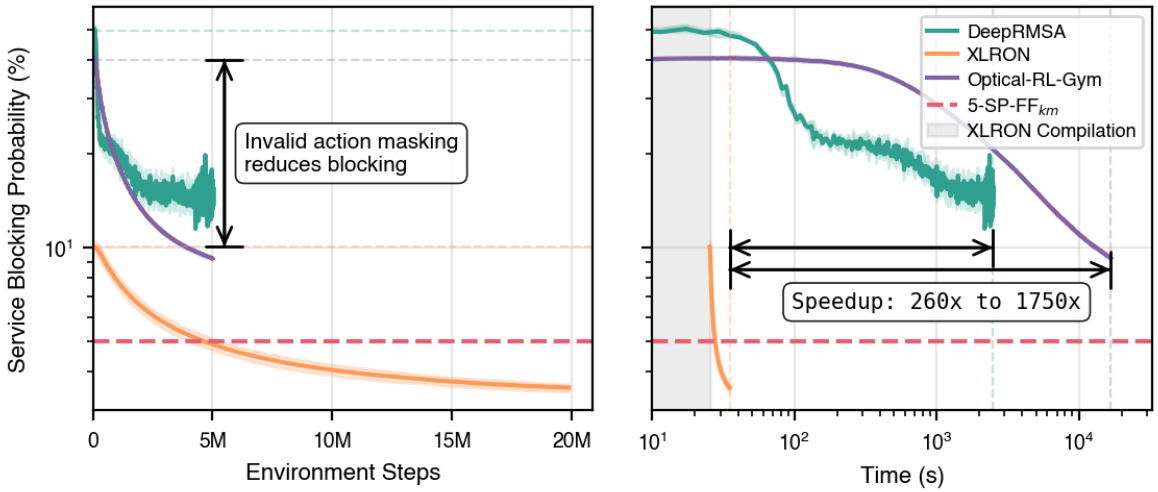


Fig. 4. RL training comparison on the DeepRMSA benchmark. Left: service blocking probability vs. environment steps. Right: the same curves against wall-clock time. DeepRMSA (green) and ORL-Gym (purple) are trained for 5×10^6 steps and XLRON (orange) for 2×10^7. The shaded region is XLRON's one-time JIT compilation, excluded from the speedup factors quoted in the text. The dashed red line is the KSP-FF heuristic with paths ordered by length (km).

3. PERFORMANCE BENCHMARKS

This section evaluates the computational performance of XLRON across a range of problem configurations and hardware settings. Unless otherwise stated, all benchmarks used XLRON v1.6.0 with jaxlib 0.9.1 on a single NVIDIA H100 GPU (GPU results) or an Apple M1 Pro 10-core CPU (CPU results, including GNPy) — a laptop processor rather than a high-core-count server — and ran on the NSFNET topology with 100 frequency slot units (FSU) per link, $k = 5$ candidate paths and the k-shortest-path first-fit (KSP-FF) heuristic, under continuous (non-episodic) operation at a fixed load of 250 Erlangs with no warm-up period; RWA-LR instead uses incremental loading, in which services do not expire. These settings follow the DeepRMSA benchmark configuration [2], which we reproduce in Section B and which subsequent RL-based RMSA work has built on [16, 17], making it a convenient common reference point; KSP-FF is used as the common heuristic because it is the one algorithm implemented across all of the libraries surveyed. Requests arrive as a Poisson process with exponentially distributed holding times and uniformly distributed source–destination pairs; each of these is configurable, including via a custom traffic matrix, for research into other traffic patterns. Every configuration is run five times and the mean reported. All throughputs were evaluated over runs long enough for execution time to dominate fixed overheads and for the network to reach steady-state occupancy, and exclude the one-time compilation cost.

A. Computational Performance

Environments benchmarked We benchmark five resource-allocation problems, implemented as separate environments. RWA (Routing and Wavelength Assignment) and RMSA (Routing, Modulation, and Spectrum Assignment) are the canonical formulations, RMSA differing in its flex-grid spectrum and distance-adaptive modulation format per request; RWA-LR (RWA with lightpath reuse) is a fixed-grid, flex-rate-transceiver variant introduced by Nevin *et al.* [28]. RSA-GN and RMSA-GN add the closed-form ISRS GN physical-layer model so that allocation decisions are SNR-aware: RSA-GN allocates a fixed spectral width per request, whereas RMSA-GN sizes each request by the highest-order modulation format its computed SNR can support and re-evaluates every active lightpath after each allocation, making it the most computationally intensive variant.

Throughput against other libraries Figure 1 compares the throughput of all surveyed libraries for dynamic allocation of 10^5 RMSA requests on NSFNET (100 FSU per link); the data source for each entry is given in the caption. Throughout this paper NSFNET denotes the 14-node variant distributed with the DeepRMSA reference implementation [2]: 22 bidirectional links, modeled as 44 directed. On a single CPU core XLRON reaches 46K SPS, exceeding the 20K SPS of Flex Net Sim, the fastest of the other libraries; parallelized across 4,096 environments on one H100 GPU it reaches 1.18×10^7 SPS, nearly 600× Flex Net Sim's single-core figure. The physical-layer-aware environments are necessarily slower per step, but even a single GN environment on GPU runs about 40× faster than ONGym's GN model on CPU and over 140× faster than GNPy.

Single environment versus parallel scaling For a single environment, the resource-allocation-only environments reach 14,300–45,700 SPS on one CPU core but only 7,400–11,400 SPS on GPU, because the GPU's cores sit idle and per-step kernel-launch latency dominates. The GN-model environments invert this, their physical-layer arithmetic being heavy enough to keep the GPU busy even at one environment: RSA-GN reaches about 5,300 SPS on GPU against 133 on CPU, and RMSA-GN about 3,200 against 20. Figure 2 shows the balance shifting with parallelism: GPU throughput scales near-linearly to 1.18×10^7 SPS at 4,096 environments, whereas CPU throughput saturates near 2.2×10^5 SPS once it exhausts its core count. The curves cross at roughly ten environments. Mixed-precision execution is used at the largest counts, both because it is faster and because it is needed to fit 4,096 environments in device memory.

Scaling across topologies Figure 3 surveys throughput across six topologies — from the 11-node COST239 network to the 75-node CONUS network [29] — and all five environments. The

six were chosen to span a range of network sizes and structural characteristics, and because each is widely used in the optical networking literature. The resource-allocation-only environments sustain several hundred thousand SPS on GPU across every topology, varying only modestly with network size. The GN-model environments are far more size-sensitive: at a single environment on GPU, RMSA-GN ranges from about 3,200 SPS on NSFNET down to 325 SPS on CONUS, and on CPU manages only 2–20 SPS — underscoring that GPU acceleration is essential for physical-layer-aware simulation at scale. At matched parallelism RSA-GN is consistently the faster of the two GN environments — by 1.7–3.1× on GPU and 6–16× on CPU — due to RMSA-GN's per-allocation SNR re-evaluation of every active lightpath.

Physical-layer cost and compilation The GN-model environments' per-step cost grows with the number of channels, so their throughput falls as the transmission band widens. For context, we benchmarked GNPy on NSFNET under the same conditions (100 random 100 Gbps demands via `path_requests_run`, default equipment library): the mean was 26.9 ms per request (range 4.3–71.5 ms). On CPU the XLRON GN environments span from about 7.5 ms per request for RSA-GN to tens of milliseconds for RMSA-GN; the former matches the 7.5 ms reported for ON-Gym's Cython-based GN model with a first-fit heuristic [5]. On GPU XLRON takes 0.2–0.3 ms per request while still computing the SNR impact of every candidate allocation. Every XLRON environment incurs a one-time JIT compilation completing in under 10 s on CPU and under 1 minute on GPU, even with hundreds of parallel environments and the GN model; this cost is amortized over the run and is not repeated when traffic load changes.

B. Reinforcement Learning Benchmarks

XLRON's greatest differentiator is its tight integration with RL training. Its implementation of the proximal policy optimization (PPO) algorithm [30] — widely used, and behind some of the most prominent successes in deep RL [31] — is written in JAX alongside the environments, so the training loop compiles together with the simulation and runs on the same accelerator, eliminating the CPU-GPU data transfer bottleneck [32].

To evaluate the XLRON end-to-end RL training performance, we reproduced the DeepRMSA benchmark from Chen *et al.* [2] and compared the training dynamics and wall-clock time of three frameworks: the original DeepRMSA implementation, Optical-RL-Gym [4], and XLRON. All three were trained on the same problem, following the original DeepRMSA configuration: RMSA on NSFNET with 100 FSU per link, k=5 candidate paths, continuous operation with a warmup period, and truncated service holding time. XLRON is trained for 2×10^7 environment steps and the other two for 5×10^6 each, their slower execution making a full-length run impractical. XLRON runs on a single NVIDIA H100 GPU with jaxlib 0.9.1; DeepRMSA and Optical-RL-Gym run on the same Apple M1 Pro 10-core CPU host, being CPU-based frameworks that do not benefit from GPU acceleration as implemented. The KSP First-Fit heuristic (5-SP-FF$_{km}$) is shown as a baseline.

Figure 4 presents the training curves. In the left panel XLRON achieves significantly lower blocking probability than both baselines throughout training, owing to its invalid action masking, which constrains the agent to feasible spectrum allocations. The right panel shows the same curves against wall-clock time. XLRON completed its full 2×10^7-step run in 9.6 s on GPU (not including a one-time compilation period of 25 s, shown in grey), whereas DeepRMSA and Optical-RL-Gym require 2,519 s and 16,795 s respectively to reach 5×10^6 steps — a quarter of the XLRON step count. Taking the ratio of each baseline total training time to XLRON's 9.6 s of execution time gives speedups of 260× and 1,750× respectively; including XLRON's compilation overhead the factors become 72× and 480×.

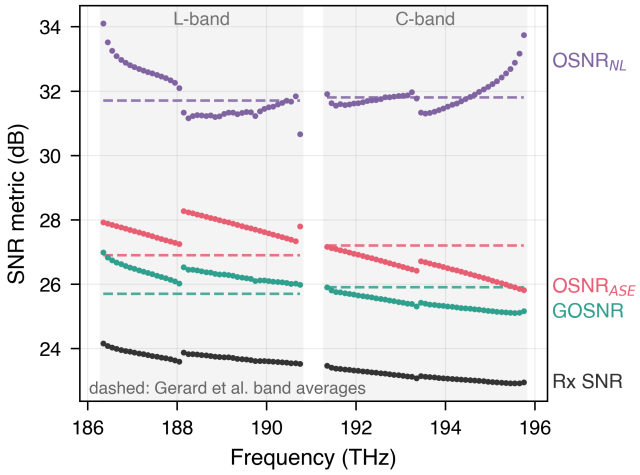


Fig. 5. Per-channel SNR metrics from the XLRON simulation (dots) compared with the per-band average values reported by Gerard *et al.* [33] (dashed lines).

4. PHYSICAL LAYER MODEL

The Gaussian noise (GN) model treats the nonlinear interference (NLI) generated by the Kerr effect as additive Gaussian noise, so that a lightpath can be characterized by an effective SNR instead of by propagating its waveform [34–37]. In its original form the NLI power is a multi-dimensional integral over the transmitted spectrum, evaluated numerically for every channel; *closed-form* variants approximate that integral with analytic expressions that are orders of magnitude cheaper to evaluate, which is what makes physical-layer-aware simulation viable inside a resource allocation loop [38–40]. XLRON implements the closed-form ISRS GN model developed over a series of papers at UCL [26, 27, 41–50], which extends the GN model to the wideband regime by accounting for inter-channel stimulated Raman scattering (ISRS), and further to arbitrary span lengths and losses, distributed Raman amplification (DRA), and a modulation-format-dependent correction. The closed form with DRA has itself been validated experimentally in an S+C+L-band hybrid-amplified long-haul system [51], and the wider context of ultra-wideband system modeling is reviewed in [52].

XLRON combines this NLI estimate with amplifier ASE and transceiver noise to report per-channel SNR. Three optical metrics are reported throughout this section, following the convention of the experimental literature: $\mathrm{OSNR}_{\mathrm{ASE}}$ and $\mathrm{OSNR}_{\mathrm{NL}}$ are the optical SNRs a channel would have were amplifier ASE or nonlinear interference respectively its only noise source, and the generalized OSNR (GOSNR) combines the two, $\mathrm{GOSNR}^{-1} = \mathrm{OSNR}_{\mathrm{ASE}}^{-1} + \mathrm{OSNR}_{\mathrm{NL}}^{-1}$, giving the single optical figure of merit that maps to received signal quality. In this section, we evaluate the accuracy of this model by recreating

the transmission system described by Gerard *et al.* [33], who demonstrated 72 Tb/s real-time transmission over 1,504 km of G.654.E fiber using hybrid Raman-EDFA amplification across the C+L bands — at the time of publication, the highest real-time capacity-distance product achieved in a multi-band single-core system. We chose this experiment because it is an exemplary high-throughput demonstration built from sophisticated techniques — hybrid Raman-EDFA amplification across C+L with real-time commercial transponders — which XLRON reproduces to good approximation. It is a point-to-point link, split into two optical multiplex sections by an intermediate pair of ROADMs; XLRON applies the same per-span model along every route of a mesh, so a network is simulated as an ensemble of such connections.

A. Experimental Reference

The system described in [33] consists of 15 spans of 100 km Corning TXF fiber (a cut-off-shifted single-mode fiber compliant with ITU-T G.654.E), each supported by a backward-pumped first-order Raman amplifier with five pumps (1,420–1,495 nm) followed by separate C- and L-band EDFAs. The C and L bands are each 4.8 THz wide with a 275 GHz inter-band gap, and 90 channels (45 per band) carry 100.4 GBd probabilistic constellation-shaped (PCS) 64-QAM signals. The per-band optimal fiber input power, found by digital-twin-based particle swarm optimization, was 18.3 dBm (C) and 18.8 dBm (L) with near-zero intra-band tilt; per-span loss was (18.4 ± 0.3) dB with average Raman gain (8.7 ± 0.6) dB. The key result is the per-band average performance in Table I of [33]: GOSNR of 25.9 dB (C) and 25.7 dB (L).

B. XLRON Simulation Setup

We configured the XLRON physical layer model to match the Gerard *et al.* system as closely as possible using the information available in [33]. All physical-layer simulations were run on an Apple M1 Pro 10-core CPU using JAX and jaxlib versions 0.9.1. The simulation uses approximately the same fiber parameters (TXF G.654.E: attenuation 0.16 dB/km, nonlinear coefficient 0.9 $W^{-1}km^{-1}$, dispersion 21 ps/nm/km, Raman gain slope 1.0×10^{-17} $W^{-1}m^{-1}Hz^{-1}$), span length (100 km) with 1.0 dB of additional lumped loss, number of spans (15), and the channel plan described above. Distributed Raman amplification is provided by five backward pumps at 200.5–211.1 THz carrying 700 mW in total, followed by per-band EDFAs with noise figures of 8.5 dB (C) and 9.0 dB (L); the hybrid noise figure follows from the standard Friis cascade rule [53]. Each channel is divided into 8×12.5 GHz Nyquist digital subchannels, improving the accuracy of the self-phase modulation contribution to the nonlinear interference [54]. The modulation-format-dependent EGN correction is disabled throughout this section, so the results below exercise the ISRS GN model alone. The Raman gain profile is obtained by integrating the coupled signal–pump equations along each span and fitting the resulting power profiles to the five-parameter closed-form ansatz of Semrau *et al.* [45]; this semi-analytical fit is cheap but leaves a small residual ripple, visible in the figures below, relative to a full numerical solution.

C. SNR Comparison

Figure 5 compares the $\mathrm{OSNR_{ASE}}$, $\mathrm{OSNR_{NL}}$, and GOSNR per channel against the per-band averages reported by Gerard *et al.* The simulated GOSNR (teal dots) is distributed closely around

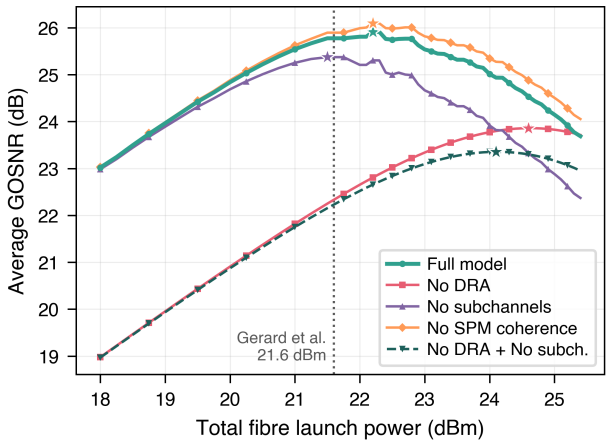


Fig. 6. Ablation study: average GOSNR vs. total fiber launch power for different model configurations. The full XLRON model (teal) is compared with the experimentally determined optimal launch power from Gerard *et al.* [33] (vertical dashed line at 21.6 dBm). Each removed model component shifts the optimum launch power and GOSNR: DRA ("No DRA"), Nyquist subchannels ("No subchannels"), or coherent multi-span SPM accumulation ("No SPM coherence").

the experimental reference for both bands (dashed lines), with frequency-dependent variation arising from SRS-induced power transfer and Raman gain. The $\mathrm{OSNR_{ASE}}$ (red dots) exhibits a within-band tilt with a superimposed sawtooth: amplifier gain is constrained to compensate span loss including the ISRS power transfer toward lower frequencies, evaluated with the closed-form solution of Zirngibl [55], so higher-frequency channels receive more gain and correspondingly more ASE noise, while the sawtooth tracks the residual ripple of the Raman-profile fit. The $\mathrm{OSNR_{NL}}$ (purple dots) varies from approximately 31 to 34 dB across the transmitted bandwidth.

Figure 5 also shows the received SNR, which is relatively flat thanks to the hybrid Raman-EDFA amplification and which adds to the optical GOSNR the per-band transceiver implementation penalty specified in the model noise-parameter file. Gerard *et al.* do not tabulate a received SNR, so this curve does not enter the quantitative comparison below; its 23–24 dB range is consistent with the Q-factors they report for their 800 Gb/s PCS-64QAM channels.

Averaging the per-channel results of Figure 5 over each band gives a direct comparison with the values reported by Gerard *et al.* The nonlinear noise estimate agrees most closely: $\mathrm{OSNR_{NL}}$ differs by 0.1 dB (C-band: 31.9 vs. 31.8 dB) and 0.2 dB (L-band: 31.9 vs. 31.7 dB). GOSNR agrees to within 0.6 dB (C-band: 25.4 vs. 25.9 dB; L-band: 26.3 vs. 25.7 dB) and $\mathrm{OSNR_{ASE}}$ to within 0.9 dB (C-band: 26.5 vs. 27.2 dB; L-band: 27.7 vs. 26.9 dB). The residual per-band offsets are anti-symmetric — the simulation slightly favors the L band where the experiment favors the C band — reflecting the amplification model: we constrained per-band gain to exactly compensate span loss under the modeled ISRS tilt, whereas the experimental pump powers and EDFA gains were independently optimized and are not fully specified in [33]. At the system level the aggregate estimate is closer still: summing the per-channel achievable information rates gives 72.4 Tb/s, within 0.6% of the 72.0 Tb/s achieved experimentally.

D. Ablation Study

To quantify the effects of the components of XLRON physical layer model (ISRS GN model with DRA, Nyquist subchannels, and coherent NLI accumulation), we perform an ablation study in which individual model features are disabled and the average GOSNR is computed as a function of total fiber launch power, a parameter whose optimum is sensitive to the fidelity of the underlying model [56, 57]. The results are shown in Figure 6, with the total launch power used by Gerard *et al.* indicated by the vertical dashed line at 21.6 dBm (the sum of their per-band fiber input powers of 18.3 and 18.8 dBm).

The full model predicts a peak average GOSNR of approximately 25.9 dB at a total launch power of approximately 22.2 dBm, within 0.6 dB of the experimental operating point. Removing the distributed Raman amplification model ("No DRA", red) shifts the optimal launch power to 24.6 dBm and reduces the peak GOSNR by approximately 2 dB, demonstrating the significant impact of DRA on both the optimal operating point and achievable performance. Using 100 GHz channels instead of dividing into 8×12.5 GHz Nyquist digital subchannels ("No subchannels", purple) reduces the peak GOSNR by approximately 0.5 dB, as the wider channels suffer from increased SPM contribution to NLI. The most degraded configuration ("No DRA + No subch.", dark teal, dashed) yields the largest deviation from the full model. The "No SPM coherence" variant (orange) replaces the coherent multi-span SPM accumulation of the ISRS GN model, in which contributions from successive spans are partially correlated and scale super-linearly with span count [34], with a strictly incoherent power summation, raising the GOSNR estimate by approximately 0.2 dB.

These results highlight the importance of DRA, Nyquist subchannel and coherent NLI modeling for accurate performance prediction in wideband C+L systems: simplified models introduce errors of 1–2 dB in GOSNR and shift the predicted optimal launch power by several dB, whereas the full model agrees with experiment to the accuracy reported in Section C.

5. DIFFERENTIABLE SIMULATION

Because XLRON simulations are fully differentiable, gradients of any simulation output — total throughput or blocking probability, say — can be computed with respect to any input parameter, continuous physical parameters and discrete resource allocation decisions alike. XLRON is, to our knowledge, the first optical network simulator to provide a fully differentiable simulation environment for resource allocation problems.

The principal challenge in applying automatic differentiation (AD) to optical network simulation is that the core operations of path selection, slot assignment, and checking action validity are inherently discrete and therefore non-differentiable. XLRON addresses this using the straight-through gradient estimator [14], which substitutes a differentiable approximation of each discrete operation during the backward pass while preserving the exact discrete computation in the forward pass. Concretely, an operation such as `argmax` is evaluated exactly on the forward pass, so the simulator state is always physically meaningful, but on the backward pass the gradient is computed as if it were a softmax with temperature τ, $\sigma_\tau(\mathbf{z})_i = \exp(z_i/\tau)/\sum_j \exp(z_j/\tau)$. Small τ sharpens the softmax toward the true `argmax` but the gradient vanishes; large τ gives a smoother surrogate with stronger but more biased gradients. XLRON provides differentiable versions of all discrete operations used in the simulation, including conditional selection, comparison, rounding, array indexing, and one-hot updates [13]. The resulting gradients correspond to a smoothed version of the true discrete objective, but can nonetheless guide optimization usefully, as in a recent application to discrete-event queueing networks [58].

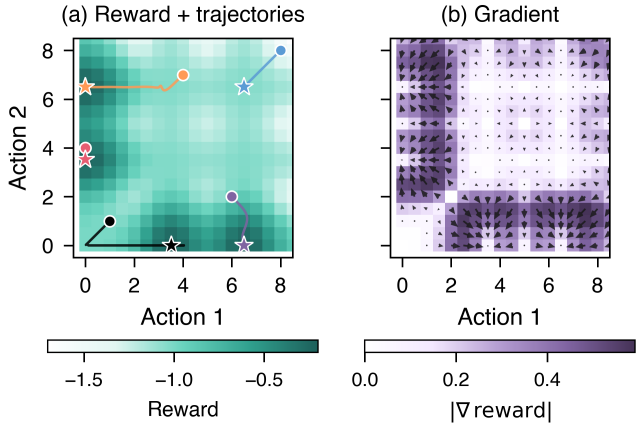


Fig. 7. Reward landscape for the two-request example. Both axes are action indices, each selecting a combined path and starting spectrum slot: Action 1 for the first request, Action 2 for the second. (a) Reward for every combination; darker shading is higher reward (both requests accommodated). Circles mark five starting points and the connected stars show where gradient-based optimization converges. (b) Gradient at each point: shading gives the magnitude and arrows point in the direction of increasing reward, toward the high-reward regions of (a).

Differentiable simulation is a promising tool for optimization in many domains, though practical implementation remains challenging [59]: chaotic dynamics produce uninformative gradients through ill-conditioned Jacobian spectra [60], and stiff or discontinuous systems compromise the bias-variance trade-off of first-order estimators relative to zeroth-order alternatives [61]. The results in this section are therefore exploratory and further research is required to overcome these potential pitfalls.

A. Reward Landscape Visualization

To build intuition for how the gradients relate to the discrete reward, we start with a deliberately small example that can be visualized in full. Two connection requests arrive one after another on a 5-node topology with three frequency slots per link and $k = 3$ candidate paths, starting from an empty network. For each request the simulator chooses a single action: an integer index that jointly selects which path to route the connection on and which spectrum slot to start it in, giving nine actions (3×3) per request. The reward is the negative of the number of blocked requests, so it takes the values 0, -1 and -2. The whole problem is therefore described by two numbers — the action taken for each request — so we can evaluate every combination and plot the reward as a two-dimensional map.

Figure 7 shows the result. The reward is flat within each cell and changes only when an action switches to a different path or slot, which gives panel (a) its blocky appearance. Panel (b) shows that, despite this, the gradients XLRON computes are strongest near the boundaries between reward levels and point

toward the high-reward regions of panel (a), supplying a useful sense of direction toward better allocations. Not every run reaches the best region — the run starting at the top right settles on a lower plateau — but this alignment between the gradients and the high-reward regions is the signal we exploit at realistic scale in the next experiment.

B. Direct RSA Optimization via Differentiable Simulation

We now evaluate the differentiable simulation on a realistic-scale problem. Rather than training a policy network, we treat the action at each time step as a continuous parameter and use first-order gradients from the differentiable environment to directly maximize the total number of accepted services.

Setup. We considered a routing and wavelength assignment (RWA) problem on the NSFNET topology (14 nodes, 44 directed links, the same DeepRMSA variant used in Section 3) with 100 FSU per link. A fixed sequence of 2,000 requests is generated under incremental loading (no service expiry), and the optimizer must find an action for each request to maximize the cumulative reward. All 2,000 actions are optimized jointly against the full sequence, so this is an offline problem in which the optimizer sees the whole traffic stream in advance, whereas the KSP-FF baseline commits to each allocation online; the question is whether approximate gradients can navigate a 2,000-dimensional discrete problem at all, not whether they beat an online heuristic on equal terms. The Adam optimizer [62] is used with a learning rate of 1.0. The straight-through temperature τ was chosen by a coarse sweep per case and is the one parameter the results are appreciably sensitive to.

Initialization strategies. We compare two initialization strategies against the KSP-FF heuristic baseline. **Zero init.** sets all 2,000 actions to zero — equivalent to taking the first slot on the first available path for every request — and runs for 20,000 iterations at $\tau = 2.7 \times 10^{-4}$, the gradients gradually incrementing each action over the course of optimization. **Heuristic init.** initializes the actions to the KSP-FF solution and runs for 1,000 iterations at $\tau = 5.0 \times 10^{-4}$.

Accepted services. Figure 8 shows the convergence trajectories for one representative set of traffic requests, on which the KSP-FF heuristic accepts 1,012 out of 2,000 requests. Because services never expire under incremental loading, the network fills monotonically until it saturates, so this count measures how much of the request sequence the topology can hold rather than a steady-state blocking probability. The zero-initialized optimizer climbs from 0 to 1,032 accepted services, surpassing the heuristic, while the heuristic-initialized optimizer transiently reaches 1,019 before drifting away. To assess robustness we repeat both experiments over five independent request streams. Zero-initialized optimization exceeds the KSP-FF baseline on four of the five streams, by +20 services on average (up to +35), using only first-order gradient information and no domain-specific knowledge. Heuristic-initialized optimization exceeds its starting solution during every one of the five runs, but only by +1 to +7 services and only transiently: the quoted figures are best-so-far values on an oscillating trajectory rather than stable solutions. The zero-initialized runs take approximately 15 minutes each on the Apple M1 Pro CPU and the heuristic-initialized runs 45 s; the former are still rising at their iteration limit, so 1,032 is a stopping point, not a converged optimum.

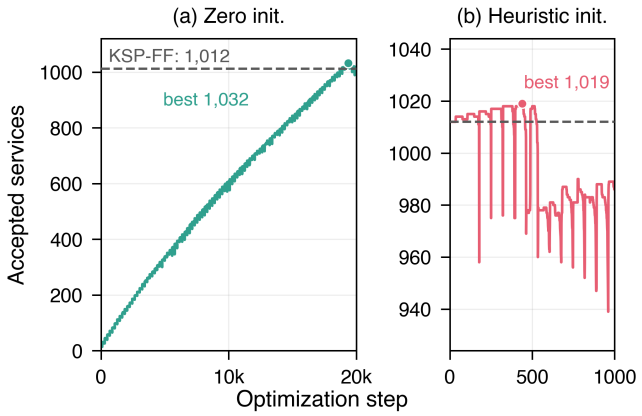


Fig. 8. Convergence of gradient-based RSA optimization on a set of 2,000 traffic requests. (a) "Zero init." starts all actions at zero and, guided only by gradients, climbs past the KSP-FF baseline (dashed gray). (b) "Heuristic init." starts from the KSP-FF solution, briefly exceeds it, and then drifts away from the optimum (note the zoomed y-range).

Discussion. These results demonstrate that the straight-through gradient estimators provide useful gradient signal for discrete resource allocation. That an optimizer starting from zero typically exceeds the KSP-FF heuristic is notable given the combinatorial difficulty of jointly optimizing 2,000 allocation decisions, each of which constrains the feasibility of all subsequent ones, and suggests that differentiable simulation can act as a standalone optimizer rather than merely refining an existing solution.

The gradients are nonetheless approximate: they guide the optimizer through a smoothed landscape that does not perfectly correspond to the discrete objective, which manifests as instability. When initialized near a good solution (Heuristic init.), the optimizer briefly exceeds the heuristic in every stream but does not hold the improvement, as the approximate gradients push actions toward regions that are favorable under the differentiable approximation but suboptimal under the true discrete evaluation. Maintaining stability near an optimum therefore remains an open challenge, and the strongest results here come from the extended from-scratch optimization rather than local refinement.

These observations are consistent with the broader literature on differentiable discrete optimization [59] and suggest avenues for future work: hybrid schemes alternating gradient-based refinement with discrete local search, and using differentiable gradients to augment RL training [63].

6. INTERFACES AND USAGE

XLRON provides a graphical user interface (GUI) for interactive use and a command-line interface (CLI) for programmatic access. Both expose the full set of configuration options for every execution mode, lowering the barrier to entry for new users while supporting advanced workflows.

The GUI (Figure 9) is a browser-based application launched with the `xlron` command, usable locally or over secure shell (SSH) from a remote server. Tabbed sections expose every configuration parameter with descriptive tooltips; users select an execution mode — RL training, heuristic evaluation, model

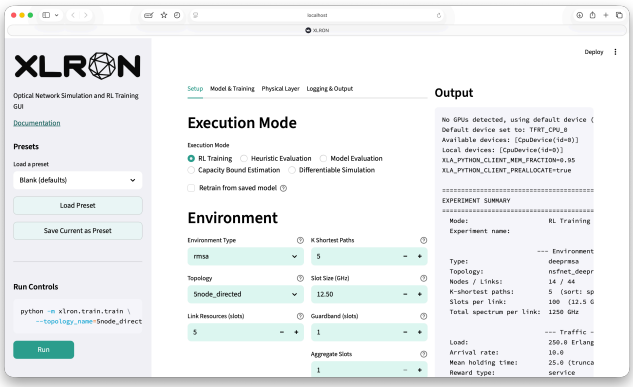


Fig. 9. The XLRON graphical user interface, showing configuration options (left and center) and real-time experiment output (right).

evaluation, capacity bound estimation or differentiable simulation — configure the environment, and launch with a single click. The left panel shows the constructed command and a "Run" button, the right panel streams the experiment summary, training progress and performance metrics, and presets can be saved and reloaded for reproducible workflows. During simulation, the network state can be visualized using XLRON's `render` method, which displays per-link spectrum allocation, the topology with the current request highlighted, and metrics such as blocking probability and utilization. It is available for all environment types and useful for debugging, demonstration, and gaining intuition about agent or heuristic behavior.

The CLI is the primary interface and the foundation on which the GUI is built. All features are accessible through command-line flags, making it straightforward to script and sweep large sets of experiments. It is intended to be used by LLM-based coding agents and humans: a single namespace with consistent parameter naming and comprehensive documentation lets an agent construct an XLRON command from a natural-language description of an experiment. The accompanying suite of over 2,200 unit tests provides the regression safety net that agent-authored extensions require.

7. CONCLUSION

This paper has presented XLRON, a unified framework for simulating dynamic resource allocation in wavelength-routed and flex-grid optical networks that combines comprehensive features with high computational performance, and shown through comparison with seven other open-source libraries that it provides the most extensive feature set among those surveyed (Table 1).

The computational performance benchmarks demonstrate throughputs of up to 1.18×10^7 steps per second (SPS) on GPU, which on the DeepRMSA benchmark translates into end-to-end RL training 260× to 1,750× faster than the reference frameworks as shipped. Even the most computationally intensive environment (RMSA with GN model and full SNR re-evaluation) achieves over 3,000 SPS in a single GPU environment, enabling physical-layer-aware simulation at scales that were previously impractical.

The accuracy of the XLRON physical layer model was validated by recreating the 72 Tb/s C+L-band transmission experiment of Gerard *et al.* [33]. The simulated per-band GOSNR agreed with the experimental measurements to within 0.6 dB (nonlinear noise to within 0.2 dB, ASE noise to within 0.9 dB), and the estimated total achievable information rate of 72.4 Tb/s matched the measured 72.0 Tb/s to within 0.6%. An ablation study confirmed the importance of including distributed Raman amplification and Nyquist subchannel modeling for accurate performance prediction in wideband systems.

XLRON's differentiable simulation — the simulator is differentiable end to end, so network objectives such as throughput can be optimized directly by gradient descent — was demonstrated on discrete resource allocation decisions: starting from scratch and using only first-order gradients, the optimizer exceeded the KSP-FF heuristic baseline on four of five request streams of an offline 2,000-request NSFNET RWA problem (by 20 services on average). Bridging the gap between the approximate smoothed gradients and the exact discrete objective, particularly maintaining stability near an optimum, nonetheless remains an open challenge.

Several limitations bound these results. The XLRON physical layer model does not currently include polarization effects or detailed component modeling, both of which GNPy models and which matter for equipment-level planning; the DRA model used in Section 4 is not yet part of the public release. JAX's requirement that array shapes be fixed at compile time means the topology, spectrum size and candidate-path count are compile-time constants, so a change to any of them triggers recompilation. Finally, the differentiable results are exploratory: the gradients are approximate by construction, and further research is required to investigate their utility in practical optimization problems.

XLRON is open source and available at `https://github.com/micdoh/XLRON`, with documentation at `https://micdoh.github.io/XLRON/`. We anticipate that XLRON's combination of high-throughput simulation, accurate physical layer modeling, automatic differentiation, and accessible interfaces will make it a valuable tool for the optical networking research community.

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